

HMS 2.0 Phase 0-2

End-to-End Test Cases and Sample Data for Admin Operations Console

Purpose: give testers a repeatable, user-focused script for validating the admin/staff HMS web application through customer, reference data, product, asset, inspection, certificate, retest schedule, reporting, and system views.

Item	Value
Application under test	Staff/Admin Operations Console
Admin URL	http://127.0.0.1:5173/
API URL	http://127.0.0.1:8000/
Inspector app	http://127.0.0.1:5174/ - separate Phase 3 field app, not part of this Phase 0-2 admin E2E script
Data safety	Use local synthetic seed data only. Do not connect this script to production HMS.

1. Scope and Tester Rules

- This document covers Phase 0, Phase 1, and Phase 2 admin/staff workflows.
- The primary end-to-end business path is: Customer -> Reference Data/Product -> Asset -> Inspection -> Approval -> Certificate -> Retest Schedule.
- Use a fresh unique suffix for each run to avoid duplicate-code conflicts.
- Treat lifecycle actions such as Archive, Supersede, and Revoke as destructive within the local test database. Run them only on records created for the test run or on disposable seeded data.
- Expected source indicator for online/backend testing is Backend, not Mock data. If the UI shows Mock data, backend is unavailable or the request fell back.

2. Environment Setup

Step	Command or Action	Why it is needed	Expected Result
1	cd backend && uv run alembic upgrade head	Applies local database schema migrations before the UI writes records.	Migration completes without errors.
2	uv run python -m hms_backend.app.tooling.local_seed	Creates synthetic customers, assets, products, inspections, certificates, and retest schedules for repeatable tests.	Summary shows counts for customers, products, assets, inspections, certificates, and schedules.
3	uv run uvicorn hms_backend.app.main:app --host 127.0.0.1 --port 8000	Starts the local API used by the staff UI.	http://127.0.0.1:8000/health returns status ok.
4	cd web/apps/staff && npm run dev -- --host 127.0.0.1 --port 5173	Starts the actual admin Operations Console.	http://127.0.0.1:5173/ loads the Dashboard.

3. Standard Test Data

Use this exact sample set, replacing 001 with a unique run suffix if needed.

Record	Field	Sample Value	Notes
Customer	Customer name	E2E Test Customer 001	Creates customer account.
Customer	Customer code	E2E001	Must be unique.
Reference Standard	Standard code	STD001	Controlled lookup data.
Reference Standard	Standard name	E2E Test Standard 001	Used to validate reference module.
Product	Product code	PROD001	Must be unique.
Product	Product name	E2E Composite Hose 001	Product catalog item.
Product	Category/Sub category	Composite hose / E2E	Displayed in Products table.
Product	Pressure rating	Rating 1 - 1000 kPa	Use Add pressure rating.
Asset	Asset number	E2E-ASSET-001	Actual hose/assembly identifier.
Asset	Customer serial number	SER-E2E-001	Customer reference.

Record	Field	Sample Value	Notes
Asset	Lifecycle status	IN_SERVICE	Expected active asset.
Asset	Next retest due	2026-09-15	Should feed retest schedule behavior.
Asset	A end/B end	Camlock M 2 inch / Flange W 2 inch	Validates A/B end editor.
Inspection	Type/result	SERVICE / PASS	Starts as draft.
Inspection	Pressure test	2250 kPa, 300 seconds, passed, leak=none	Validates pressure result capture.
Certificate	Certificate number	CERT-E2E-001	Created from approved inspection.
Certificate	Valid until	2027-09-15	Expected expiry date.
Retest Schedule	Update values	2026-10-15, UPCOMING, 45 days, 10 days	Validates schedule editing.

4. Phase 0-2 End-to-End Test Cases

E2E-00 - Application opens against backend

Why this test exists: This proves testers are validating the real local API path instead of mock fallback data.

Steps:

- Open `http://127.0.0.1:5173/`.
- Verify the page title is BAT Engineering HMS 2.0.
- Open Live Environment in the top bar.
- Confirm the source label or popover indicates Backend connection active.

Sample data:

- No data entry.

Expected result:

- Dashboard renders without a framework error overlay.
- The UI shows operational cards such as Total Assets, In Service, Overdue, and Pending Review.
- Console has no relevant app errors. API calls should return from `localhost:8000`.

E2E-01 - Create customer

Why this test exists: A customer is the account that owns assets, contacts, locations, inspections, and certificates.

Steps:

- Click Customers.
- Click Add Customer.
- Enter Customer name, Customer code, Default retest interval, and leave Enable retest scheduling checked.
- Click Save customer.

Sample data:

- Customer name: E2E Test Customer 001
- Customer code: E2E001
- Default retest interval: 12 months

Expected result:

- Customer list count increases by one.
- New customer appears in Customer Management.
- Customer detail panel opens or can be selected, showing zero assets and zero inspections initially.

E2E-02 - Search and review customer detail

Why this test exists: This checks customer search, selection, and read-only summary panels used by admin staff.

Steps:

- Use Search customers and enter E2E001 or E2E Test Customer 001.
- Select the created customer card.
- Open Overview, Locations, Contacts, Documents, and Notes tabs.

Sample data:

- Search text: E2E001

Expected result:

- Only matching customer rows/cards remain visible.
- Detail panel shows customer ID, status, risk, metrics, and tabs.
- Tabs switch without blank content or runtime errors.

E2E-03 - Create reference standard

Why this test exists: Reference standards are controlled master data used for compliance and product classification.

Steps:

- Click Reference Data.
- Click Add Standard.
- Enter Standard code and Standard name.
- Click Save standard.

Sample data:

- Standard code: STD001
- Standard name: E2E Test Standard 001

Expected result:

- Reference Data table shows the new standard.
- Edit and Archive actions are visible for the row.

E2E-04 - Create product with pressure rating

Why this test exists: Products define hose catalog items and expected pressure matrix behavior before assets are created.

Steps:

- Click Products.
- Click Add Product.
- Enter product code, product name, category, and sub category.
- Click Add pressure rating.
- Click Save product.

Sample data:

- Product code: PROD001
- Product name: E2E Composite Hose 001
- Category: Composite hose
- Sub category: E2E

Expected result:

- Products table shows the new product.
- Pressure Matrix area accepts at least one rating row before save.
- Edit and Archive actions are visible for the row.

E2E-05 - Create asset

Why this test exists: An asset is the actual hose assembly record that inspections and certificates attach to.

Steps:

- Click Assets.
- Click Add Asset.
- Enter asset number and customer serial number.
- Select customer and product.
- Select Lifecycle status.
- Enter Next retest due.
- Enter A end and B end fitting/size values.
- Click Save asset.

Sample data:

- Asset number: E2E-ASSET-001
- Customer serial number: SER-E2E-001
- Customer: E2E Test Customer 001 if available; otherwise use seeded Vopak until dropdown refresh is fixed.
- Product: E2E Composite Hose 001 if available; otherwise use seeded FUELFLEX GREEN until dropdown refresh is fixed.
- Next retest due: 2026-09-15
- A end: Camlock M / 2 inch
- B end: Flange W / 2 inch

Expected result:

- Asset Register shows the new asset number.
- Customer, product, serial, lifecycle, retest due, and A/B ends display correctly.
- Open/Edit actions are available for the new asset.

E2E-06 - Edit asset

Why this test exists: This validates that asset maintenance fields can be changed after initial creation.

Steps:

- In Asset Register, click Edit for the test asset.
- Change serial number, lifecycle status, next retest due, and one A/B end value.
- Click Save asset.

Sample data:

- Updated serial: SER-E2E-001-REV1
- Lifecycle status: DUE
- Next retest due: 2026-10-15

Expected result:

- Asset row updates in the table.
- No duplicate asset row is created.
- Updated values persist after navigating away and back.

E2E-07 - Create inspection with pressure test

Why this test exists: Inspections capture the pressure test results that drive approval and certificate workflows.

Steps:

- Click Inspections.
- Click Add Inspection.
- Select the test asset.
- Select Inspection type Service.
- Select Result Pass.
- Enter Applied pressure kPa, Hold time seconds, Pressure test passed, and Measurement notes.
- Click Save inspection.

Sample data:

- Applied pressure kPa: 2250
- Hold time seconds: 300
- Measurement notes: leak=none

Expected result:

- Inspection table shows the test asset.
- Status is DRAFT.
- Pressure Test column shows 2250 kPa / Pass.
- Inspection detail panel opens with editable draft fields.

E2E-08 - Submit and approve inspection

Why this test exists: This validates the Phase 2 inspection status flow from draft to reviewer approval.

Steps:

- Open the draft inspection detail.
- Optionally change pressure or result fields and click Save draft.
- Click Submit inspection.
- Verify status changes to SUBMITTED.
- Click Approve inspection.

Sample data:

- Use the inspection created in E2E-07.

Expected result:

- Status progresses DRAFT -> SUBMITTED -> APPROVED.
- Reviewer and approved timestamp appear in the detail panel.
- Approved inspection becomes eligible for certificate issue.

E2E-09 - Issue certificate from approved inspection

Why this test exists: Certificates are the compliance output generated from approved inspections.

Steps:

- Click Certificates.
- Click Issue Certificate.
- Select the approved inspection for the test asset.
- Enter Certificate number and Valid until.
- Click Issue certificate.

Sample data:

- Certificate number: CERT-E2E-001
- Valid until: 2027-09-15

Expected result:

- Certificate Management table shows the certificate.
- Status is ISSUED.
- Certificate detail panel shows asset, customer, source inspection, public token, verification hash, and PDF object key.

E2E-10 - Certificate lifecycle actions

Why this test exists: This validates post-issue lifecycle controls used when certificates are replaced or invalidated.

Steps:

- Open a certificate created for the test run.
- Click Mark superseded and confirm the status changes to SUPERSEDED.
- For a separate disposable issued certificate, click Revoke certificate and confirm status changes to REVOKED.

Sample data:

- Use only local disposable certificates.

Expected result:

- Superseded certificate is no longer shown as ISSUED.
- Revoked certificate is counted in the Revoked metric.
- Lifecycle buttons disappear when no longer valid for the current status.

E2E-11 - Retest schedule update

Why this test exists: Retest schedules drive upcoming service planning and escalation workflows.

Steps:

- Click Retest Schedule.
- Open a schedule row.
- Change Retest due date, Retest status, Reminder interval days, and Escalation interval days.
- Click Save schedule.

Sample data:

- Retest due date: 2026-10-15
- Status: UPCOMING
- Reminder: 45 days
- Escalation: 10 days

Expected result:

- Schedule status updates in table and detail panel.
- Reminder and escalation intervals update.
- Due date should update and persist after navigation.

E2E-12 - Analytics and system workspaces

Why this test exists: These views validate that reporting and administration surfaces render without blocking the operational flow.

Steps:

- Open Analytics and verify fleet metrics, certificate coverage, retest risk, and inspection throughput.
- Open Users & Roles and verify role cards and access matrix.
- Open Devices and verify registered devices and offline/sync indicators.
- Open Audit Log and verify read-only activity rows.

Sample data:

- No data entry.

Expected result:

- Each workspace renders a meaningful screen.
- No blank page, framework overlay, or console runtime error appears.

E2E-13 - Topbar and utility controls

Why this test exists: These are common controls used by staff while moving between workflows.

Steps:

- Open Live Environment popover.
- Open Notifications popover.
- Open Help popover.
- Open User menu.
- Use global search with a module name such as Assets, Customers, or Certificates.
- Click New Asset and confirm it navigates to Asset Register.

Sample data:

- Global search text: Assets

Expected result:

- Each popover opens and closes without layout breakage.
- Global search navigates to the matching module.
- New Asset navigates to Assets.

E2E-14 - Filters, tabs, and download buttons

Why this test exists: These controls are used for daily review queues and record exports.

Steps:

- In Inspections, click All, Submitted, Approved, Rejected, and Draft tabs.
- In Certificates, click All Certificates, Pending Issue, Superseded, Revoked, and Issued tabs.
- In Retest Schedule, click All, Overdue, Due, Upcoming, and Suspended tabs.
- Use search boxes and source selectors in each module.
- Click download/export buttons where visible.

Sample data:

- Search text: E2E or seeded asset number 997950

Expected result:

- Tabs filter table rows and update counts.
- Search narrows visible rows.
- Download/export control should trigger expected browser download behavior or a clearly implemented action.

5. Button and Control Checklist

Area	Controls to Test	Expected Behavior
Sidebar	Dashboard, Analytics, Assets, Inspections, Certificates, Retest Schedule, Sync Queue, Customers, Products, Reference Data, Users & Roles, Devices, Audit Log	Each button opens the matching workspace and highlights active navigation.
Topbar	Global search, Live Environment, New Asset, Notifications, Help, User menu	Search navigates; popovers open; New Asset navigates to Asset Register.
Customers	Search customers, More Filters, Add Customer, detail tabs	Filtering and selection update customer list/detail content.
Reference Data	Add Standard, Edit, Archive, Filters, Download	Create/edit actions persist; archive removes local disposable records.
Products	Add Product, Add pressure rating, Save product, Edit, Archive	Product catalog rows persist and pressure matrix accepts rows.
Assets	Add Asset, Edit, Archive, Save asset, Cancel, Close form	Asset fields persist; cancel/close leave record unchanged.
Inspections	Add Inspection, Save inspection, Open, Save draft, Submit inspection, Approve inspection, status tabs	Inspection state progresses and filtering reflects current status.
Certificates	Issue Certificate, Open, Mark superseded, Revoke certificate, certificate tabs	Certificate status changes and detail panel reflect lifecycle.
Retest Schedule	Open Schedule, Save schedule, status tabs	Schedule status, reminder, escalation, and due date persist.

6. Negative and Boundary Checks

- Required fields: try saving Customer, Product, Asset, Inspection, and Certificate drawers with required fields blank. Browser validation or clear UI feedback should prevent save.
- Duplicate codes: try reusing Customer code, Product code, Standard code, Asset number, and Certificate number. The UI should show a failure state or prevent duplicate persistence.
- Backend down: stop the API and reload the staff UI. The source should switch to Mock data or clearly indicate fallback, so testers do not confuse fallback with online testing.
- Viewport: test default desktop size and a shorter 1280 x 720 viewport. Drawer actions must remain reachable without requiring hidden/off-screen clicks.
- Reload persistence: after creating records, reload the page and verify backend-backed records remain visible.

7. Known QA Watchpoints From Current Build

Watchpoint	What tester should verify	Expected Product Behavior
Asset dropdown freshness	After creating a customer/product, open Add Asset and check whether those records appear in Customer/Product dropdowns.	Dropdowns should list all active customers and products, not only customers/products already attached to existing assets.
Long drawer reachability	At 1280 x 720, open Add Product and Add Asset drawers and try to click Save without changing viewport.	Save/Cancel actions should remain reachable through scrolling or sticky drawer actions.

Watchpoint	What tester should verify	Expected Product Behavior
Asset field persistence	After saving asset with Next retest due and B end values, confirm table/detail display the saved date and both A/B ends.	All entered asset fields should persist and display after save/reload.
Certificate valid-until	Issue certificate with explicit Valid until date and verify table/detail display that exact date.	Certificate should preserve tester-entered valid-until value.
Retest due date update	Edit schedule Retest due date plus status/reminder/escalation and save.	Due date, status, reminder, and escalation should all update together.
Static dashboard counters	Create records and compare dashboard/sidebar counters.	Counters should either update from backend data or clearly show demo/static status.

8. Evidence To Capture For Each Run

- Screenshot of Dashboard showing Backend source.
- Screenshot of created customer in Customer Management.
- Screenshot of created standard and product rows.
- Screenshot of created asset in Asset Register.
- Screenshot of draft inspection with pressure test.
- Screenshot of approved inspection detail.
- Screenshot of issued certificate detail.
- Screenshot of retest schedule before and after update.
- Browser console export or note confirming no relevant errors.
- Optional API verification with curl against localhost:8000 for created records.